Vaibhav Sijaria

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Education

Vellore Institute of Technology, Vellore — **August 2023 – Present**
Bachelor of Technology in Computer Science — ***CGPA** 8.68*

Skills

Programming Languages: Python, Golang, Rust, TypeScript, SQL, Java, C, C++, Zig
Frameworks & Libraries: NumPy, PyTorch, Flask, Django, Nest.js, Prisma, Drizzle
APIs & Protocols: REST, GraphQL, tRPC, gRPC, WebSockets
Databases: PostgreSQL, MongoDB, Redis, DuckDB
Tools & Technologies: Git, Docker, Linux, GitHub Actions

Experience

Xlancr Infotech — **September 2024 – February 2025**
Software Engineer Intern — *Vellore, Tamil Nadu*

- Architected a scalable, type-safe **NestJS/TypeScript** backend with **Drizzle** handling **5,000+ RPM**; refactored it via **Dependency Injection** to enforce Separation of Concerns and improve maintainability.
- Reduced API costs and latency by ∼**50%** via a multi-layer **Redis/FastAPI** caching solution for a Python backend; enforced **70%** test coverage through a **GitHub Actions CI/CD** pipeline.

Projects

PeerWatch | *Serverless peer-to-peer video watch party* | GitHub — **August 2026**

- Built a serverless **peer-to-peer** watch-party in **Go** (stdlib only) over a full-mesh **TCP** topology, with a custom binary wire protocol (10 message types), **512KB SHA-256**-verified chunks, and sparse-file storage.
- Engineered a concurrent scheduler with duplex goroutines per peer, rarest-first prioritization, speed/load-based peer selection, request timeouts, and 3-tier drift correction over a local **HTTP Range** server to **mpv**.
- **Tech Stack:** Go, TCP, WebRTC, mpv

Competitive Coding Platform | *Real-time code submission, evaluation, and leaderboard* | GitHub — **September 2024**

- Achieved **100% uptime** while serving **600+ users** in a live competition. Implemented **Redis** caching for the duplicate submissions and contest questions, significantly reducing direct database queries and code judge load.
- Architected a secure and scalable code evaluation pipeline by integrating **Judge0** for sandboxed execution and **Asyncq** for asynchronous job queuing, ensuring fast, non-blocking submission processing.
- **Tech Stack:** Golang, Docker, PostgreSQL, Redis, Asyncq, SQLC, Judge0, Gin framework

Distributed MapReduce Framework | *Language-agnostic MapReduce engine* | GitHub — **May 2026**

- Designed a **language-agnostic** execution model that compiles user-supplied Map/Reduce source and runs it via a `stdin/stdout` key-value contract, with in-memory partition buffers that spill to disk past a threshold.
- Built crash-safe recovery via periodic **JSON checkpointing** of job/task state to S3, ephemeral worker re-registration on heartbeat loss, and **atomic output promotion** via S3 `CopyObject/DeleteObject`.
- **Tech Stack:** Golang, gRPC, Protocol Buffers, AWS S3, Docker, HTTP

Leadership & Extracurricular

Tech Coordinator, Gravitas 2025 — **September 2025**

- Engineered high-throughput payment workflow using **Go**, **PostgreSQL**, and a **Redis**-based queue system. Successfully handled **3 million+ visits** and processed over **50 lakhs** in transactions with **zero downtime** during peak load.
- Eliminated booking related **race conditions** via a 10-minute seat-hold system based on **Redis Sorted Sets**.
- Resolved a critical transaction leak on all code paths, preventing **connection pool exhaustion**; also shipped new accommodation/check-in workflows and built admin dashboards, which reduced support tickets by **95%**.

Event Sub-Coordinator, CookOff 9.0 - Gravitas 2024 — **September 2024**

- Led a 5-member backend team in developing the submission portal for **Cook Off 9.0**, which processed **15,000+ evaluations** with **zero downtime**, a peak throughput of **200+ submissions/min**, and an average verdict time of **<10s**.

1st Runner up, ForkThis - Gravitas 2023 — **September 2023**

- Contributed to open-source projects by resolving bugs and implementing features across multiple repositories. Secured **2nd place out of 300+ participants** with **25 merged PRs**, utilized **Python, JavaScript, Selenium,** and **Node.js**.